

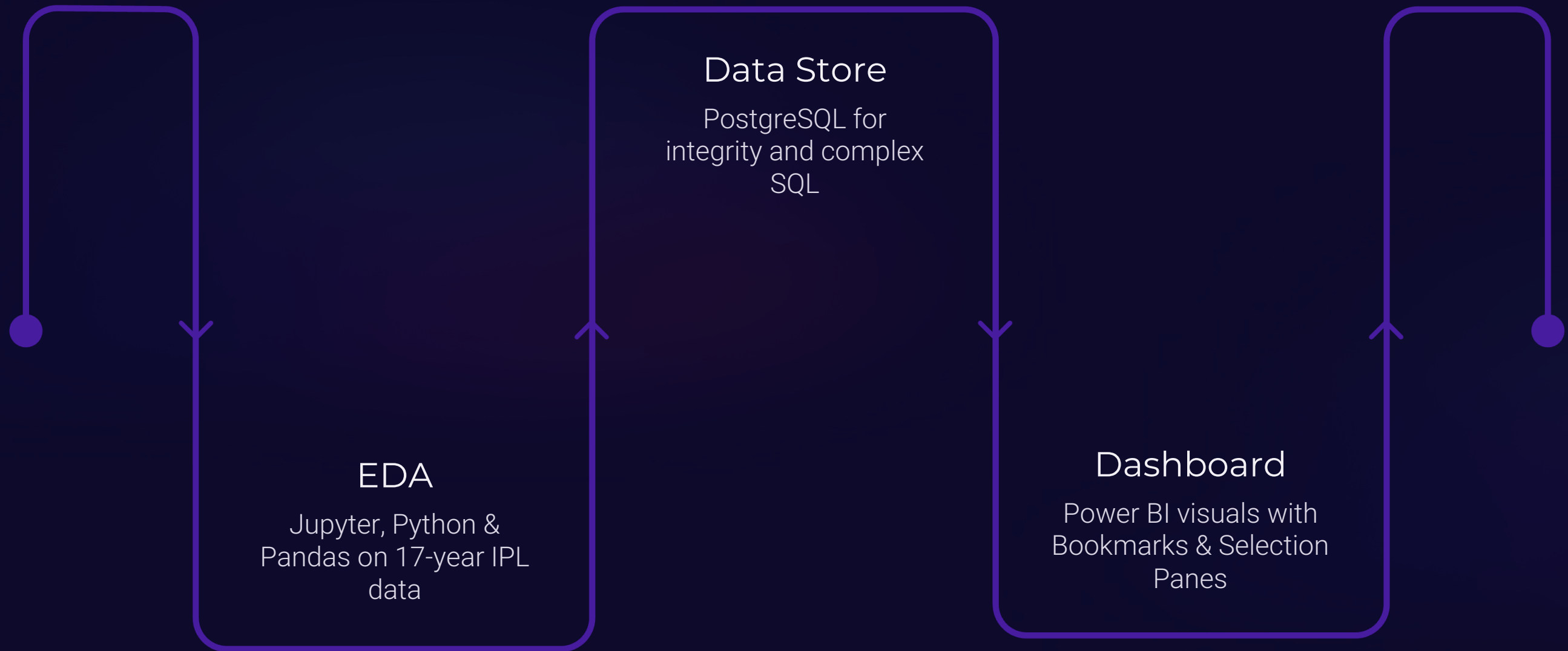


IPL 2026: Aggressive XI Scouting & Performance Analytics

A data-driven framework combining **Situational Impact** and **Venue-Specific Adaptability** to identify high-value players for the IPL 2026 auction – moving beyond reputation-based selection.

BY DEEPANSHU

A Three-Stage Analytical Pipeline



Each phase builds on the last — from raw ball-by-ball data to a fully interactive scouting dashboard ready for auction decisions.



Batsman Scouting Dashboard

The **BATSMAN** view features player Strike Rate & Average Runs, Performance by City, Boundary Percentage (Four %, Six %, Other%), and the **Batsman Aggression Matrix** — filterable by Season, Match Phase, and Innings.

Elite Batsmen by Strike Rate

Striker	Strike Rate	Avg Runs
TM Head	251.11	45.20
J Fraser-McGurk	251.06	39.33
Abhishek Sharma	222.00	22.20
SP Narine	178.89	26.83
PD Salt	177.11	24.50
RG Sharma	168.09	15.80
DA Warner	163.89	29.50



Key Insight: TM Head and J Fraser-McGurk lead with Strike Rates exceeding 251, combining elite aggression with solid averages – prime targets for the 2026 auction.

Players filtered for minimum balls faced across 2022–2024 seasons to ensure statistical significance.



Bowler Scouting Dashboard

The BOWLER view tracks Economy Rate and Wickets, Bowling Impact by City, Dot Ball Percentage out of total balls, and the Bowler Lethality Matrix — comparing Economy vs. Bowling Strike Rate across phases.

Technology Stack

Multi-Tool Architecture



Jupyter Notebook

Python 3.x with Pandas, NumPy, and Matplotlib/Seaborn for EDA, data cleaning, and feature engineering.



PostgreSQL

pgAdmin 4 as RDBMS for structured querying, data integrity, and complex aggregations on delivery-level records.



Power BI Desktop

DAX measures for custom metrics; Bookmarks & Selection Panes for a clutter-free, interactive UI.

```
import pandas as pd
from sqlalchemy import create_engine, URL

# 1. Load the RAW file
df = pd.read_csv(r"C:\Users\deepa\Downloads\archive (2)\ipl_full_info(2008-2024).csv")

# 2. THE CLEANING (Adding the missing columns)
df = df[df['year'] >= 2022].copy() # Filter for active years
df['runs_on_ball'] = df['batsman_run'] + df['extras'] # Create the missing column
df['match_phase'] = df['over'].apply(lambda x: 'Powerplay' if x < 6 else 'Middle Overs' if x < 15 else 'Death Overs')

# 3. Create the connection
connection_url = URL.create(
    drivername="postgresql+psycopg2",
    username="postgres",
    password="Deepanshu@123",
    host="localhost",
    port="5432",
    database="IPL_Auction_2026"
)

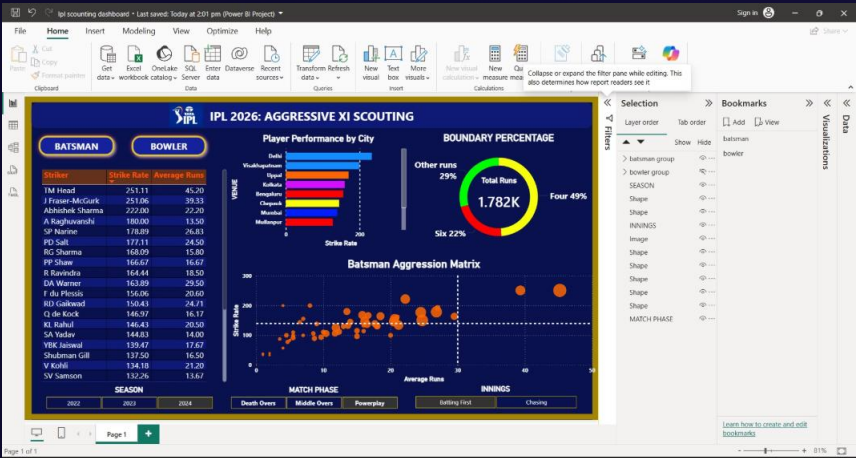
engine = create_engine(connection_url)

# 4. Push to SQL (We use 'replace' to fix the old table)
df.to_sql('ipl_full_data', engine, if_exists="replace", index=False, chunksize=10000)

print("SUCCESS! The 'runs_on_ball' column is now in pgAdmin.")
```

```
--Choosing the best Openers

SELECT
    "Striker",
    SUM(runs_on_ball) as pp_runs,
    COUNT(*) as balls_faced,
    ROUND((SUM(runs_on_ball)::numeric / COUNT(*) * 100, 2) as strike_rate
FROM ipl_full_data
WHERE match_phase = 'Powerplay'
GROUP BY "Striker"
HAVING COUNT(*) >= 100
ORDER BY strike_rate DESC
LIMIT 10;
```



From Raw Data to Actionable Insight

Data Source

Ball-by-ball IPL records, 2008–2024 (17 seasons). Captures runs, wickets, extras, venue, and match phase.

Step 1 — Pre-processing

Pandas cleaning: removed duplicates, handled nulls, engineered "Dot Balls" and "Strike Rate per Phase."

Step 2 — Database

Exported to PostgreSQL; SQL queries validated data and performed aggregations before BI connection.

Step 3 — DAX Metrics

Power BI DAX created dynamic measures: Dot Ball %, Lethality Index, Aggression Matrix, boundary % and Venue-specific Strike Rates.

Step 4 — UI/UX

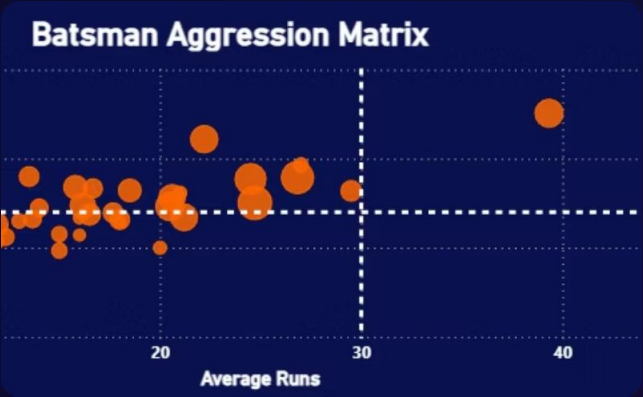
Selection Groups and Bookmarks deliver a "Single-Page App" feel — toggle between Batsman and Bowler views seamlessly.

Scope

Filtered to 2022–2024 for recency. Minimum balls faced/overs bowled thresholds ensure statistical validity.

Dashboard Visuals

Key Analytical Charts



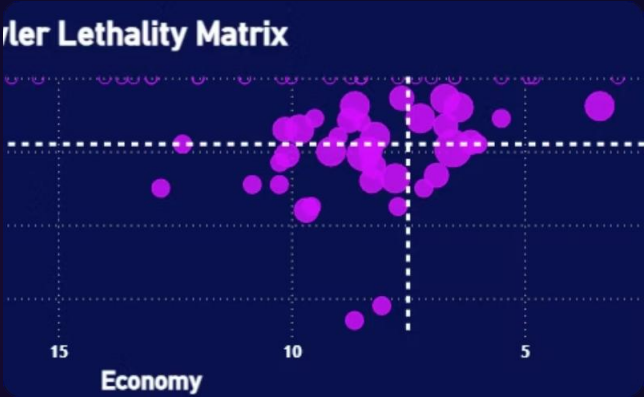
Aggression Matrix

Upper-right quadrant = "Elite Aggressors" — high Strike Rate with consistent Average Runs.



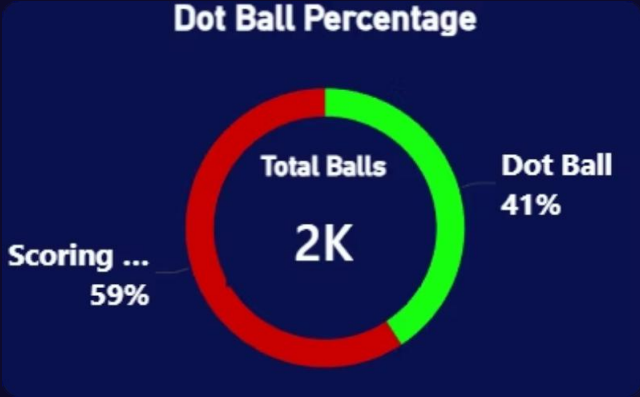
Venue Performance

Ground-specific Strike Rates across Delhi, Chepauk, Wankhede, Eden Gardens, and more.



Lethality Matrix

Economy vs. Bowling Strike Rate — identifies wicket-takers who also contain runs in Powerplay and Death Overs.



Dot Ball Analytics

Custom DAX measure: 41% Dot Ball % out of 2K balls — a critical pressure metric for bowler evaluation.

Scouting Strategies for IPL 2026 Auction

1

High-Impact All-Rounders

Target players in the top-right quadrant of *both* Aggression and Lethality matrices for squad depth.

2

Venue-Specific Specialists

For spin-friendly tracks like **Chepauk**, prioritise bowlers with high Dot Ball % and low economy in Middle Overs.

3

Age-Agnostic Aggression

Experienced players maintaining high Strike Rates in Death Overs offer tactical maturity — don't overlook them.

4

Data-Backed Risk Mitigation

Use Innings and Match Phase slicers to identify **Chasing Specialists** for high-pressure run chases.



Conclusion

The Future of Cricket Scouting is Data-Driven

Seamless Pipeline

Jupyter → PostgreSQL → Power BI:
accurate, scalable insights from raw
delivery-level data.

360° Player View

Scatter plots and donut charts deliver
multi-dimensional analysis —
aggression, consistency, and
pressure-handling.

Actionable Intelligence

A strategic asset for team
management enabling data-backed
decisions at the IPL auction table.

- ✓ This framework proves that the future of cricket scouting lies in processing historical data through a lens of **situational context** — practical for real-world sports franchises.